RF616B Bluetooth Air Mouse Remote Control Specification

Clean English-only version rebuilt from the supplier specification. Chinese-language cover pages, duplicated Chinese text, and Chinese-only sections have been removed or rewritten in English.



Front view



Side view



Back view

Field	Clean English Specification Value
Product name	Bluetooth Air Mouse Remote Control Kit
Product model	RF616B
Supplier part / model	A121RF616B0001
Customer part number	101B ONEVIEW / A0341
Supplier / manufacturer	Shenzhen C&D; Electronics Co., Ltd.
Chip program	HB9261
Firmware version	1.0.6
Revision / date	V7 / 2026-06-11
Net weight	93 g +/- 3 g
Dimensions	190 mm x 46 mm x 25.5 mm +/- 2 mm
Battery	2 x AAA alkaline batteries, shipped without batteries
RoHS	Yes

Source pages in the supplier file contain Chinese and bilingual content. This version keeps the relevant English specification content and translated technical requirements only.

1. Revision Summary

Version	Change summary	Date
V1	Initial release	2026-02-07
V2	Packaging data added	2026-04-22
V3	ID drawing updated; bottom shell laser engraving added	2026-04-27
V4	Dongle drawing added	2026-05-22
V5	Packaging drawing updated	2026-06-01
V6	Dynamic current tested after turning off backlight; packaging drawing updated	2026-06-04
V7	Static current range updated	2026-06-11

2. Product and Material Requirements

Item	Requirement
PCB	Double-sided FR4 PCB, 1.2 mm +/- 0.1 mm thickness, copper foil thickness greater than 25 um, OSP + carbon oil
PCB copper foil	Copper foil thickness greater than 25 um
IR transmitter chip	Chip size not less than 12 mil
Conductive key feel coefficient	k > 20 percent; sealed approved sample controls special cases
Carbon resistance	500 g pressure on 0.5 mm x 0.5 mm grid test board; carbon particle resistance < 100 ohm; carbon oil resistance < 200 ohm
Housing material	ABS with up to 10 percent recycled gate material
Product size	190 mm x 46 mm x 25.5 mm +/- 2 mm
Surface treatment	Raw material finish with mold texturing
Key structure	All metal dome contact structure; silicone rubber keypad
Nameplate	None

3. Electrical Performance Requirements

Test item	Requirement / result
IR peak irradiance	Under 200 +/- 50 lx ambient light at rated voltage: straight IR Power >= 40 uW/cm2; horizontal left/right 30 degrees IR Power >= 20 uW/cm2 at 0.5 m
IR remote distance	Straight distance >= 10 m; horizontal +/-30 degrees >= 5 m; vertical +/-45 degrees >= 5 m
RF remote distance	Bluetooth unobstructed key distance >= 10 m; 360-degree Bluetooth reception
Dynamic current	Backlight off test: Bluetooth <= 30 mA; IR <= 45 mA. Measured: Bluetooth 28.26 mA; IR 41.86 mA
Static current	0 to 7 uA. Measured: 4.4 uA
Operating voltage	Functions normally from 2.4 V to 3.6 V, including learning and remote operation
Key battery life	500,000 cycles at 100 presses per minute and 300 g pressure; product must function normally afterward
ESD discharge	Air discharge +/-8 kV; contact discharge +/-6 kV; capacitor 150 pF; resistor 330 ohm; no functional or component damage
EM / immunity reference	Refer to EN55013, EN61000-4-2, and EN61000-4-3
IR carrier frequency	38 kHz +/- 500 Hz; 36 kHz +/- 500 Hz
Reverse battery insertion	After 1 minute reversed battery insertion, product must work normally after batteries are reinstalled correctly; reverse test shall not conduct

4. Hardware Test Summary

Test	Method / requirement	Measured result
Crystal frequency offset	Probe line between crystal and chip; requirement < 10 ppm	4.8 ppm
RF transmit power / carrier offset	3 V DC adjustable supply; PCBA enters RF fixed-frequency transmit state; center frequency 2.440 GHz +/- 30 kHz	Spectrum analyzer test
Power ripple	Hold voice key and measure supply pin ripple; VPP < 150 mV for Bluetooth	NA in source
Battery operating life	Based on 8 h active use, 16 h standby, 300 key operations/day, and 10 min voice/day using 2 x 700 mAh AAA cells	Approx. 9.38 months; requirement > 7 months
Dynamic power	IR mode and Bluetooth connected mode, maximum current measured on OK key	IR 41.86 mA; Bluetooth 28.26 mA
Static current	No-connection state; requirement 0 to 7 uA	4.4 uA

5. Mechanical Performance Requirements

Category	Test condition / requirement
Appearance	Inspect all exterior parts visually from 30 cm in front of the product according to remote quality requirements
Alcohol resistance	500 g test rod wrapped with lint-free cloth, 95 percent alcohol, 40 cycles/min, approx. 20 mm stroke; no fading, peeling, or base exposure
Pencil hardness	750 g pencil hardness tester at 45 degrees; draw 5 lines over 5 mm; no coating damage after eraser cleaning. Standard paint HB; UV paint 2H
Eraser abrasion	FABER-CASTELL 7017R eraser under 500 g, 40 cycles/min, approx. 20 mm stroke; no obvious fading or base exposure
3M tape test	3M600 tape fully applied to printed area and peeled at 45 degrees within 1 second; 3 pulls with new tape each time
Cosmetic resistance	Apply Nivea cream to 15 mm x 15 mm printed area for 24 h; print must remain clear, legible, and not peeled
Artificial sweat	Product sealed with wetted filter paper and held at 40 +/- 2 deg C for 48 h; no staining, corrosion, or delamination
Cross-hatch	1 mm grid, 100 squares, 3M600 tape pull at 45 degrees; coating detachment area <= 5 percent
Key operating force	4 mm diameter test head pressed vertically at key center; all keys 240 +/- 100 g
Key rebound	Keys shall not jam in the top cover
Battery cover life	Battery cover opened and closed 100 times; no abnormal change in open/close force
Spring force	Negative spring rebound force shall not be less than 3 N at minimum compressed battery height
Drop test	Remote with batteries dropped from 80 cm on 6 faces and 4 corners onto wood floor at least 3 cm thick; product must function normally
Packaging vibration	Packed carton tested at 10 Hz - 30 Hz - 10 Hz, amplitude 1.5 mm, vertical direction, 2 h; no functional failure
Packaging drop	Packed carton dropped from 60 cm; test three faces, one edge, and one corner; no functional failure

6. Environmental and Packaging Requirements

Test	Condition	Requirement
High-temperature storage	60 deg C, 48 h, 5 pcs	After 1 h at room temperature, no functional or appearance issue
Low-temperature storage	-25 deg C, 48 h, 5 pcs	After 1 h at room temperature, no functional or appearance issue
Constant damp heat storage	55 deg C, 93 +/- 3 percent RH, 96 h, 5 pcs	Electrical and mechanical function must meet specification
Thermal shock storage	60 deg C for 2 h, -30 deg C for 2 h, transfer time 3 min, 5 cycles	No functional loss after 1 h at room temperature; coating shall not peel, crack, discolor, or show abnormal appearance
High-temperature aging	Remote with batteries placed in 45 deg C chamber for 96 h; function checked every 24 h	Electrical parameters shall comply; no deformation, discoloration, mechanical, or appearance abnormality
Salt spray	35 +/- 2 deg C, 85 percent RH, 5 percent NaCl, pH 6.5 to 7.2, 48 h exposure plus 48 h drying	Metal parts shall not rust except acceptable cut-edge rust
Packaging transport drop	Entire carton dropped from 60 cm on three axes, one edge, and one corner	No functional failure
PE bag	PE bag must meet environmental requirements and include recycling mark printing	Barcode control required
RoHS	Routine test requirement	Must meet environmental compliance requirements

7. Cleaned Product Visual Reference

The original drawing pages include Chinese labels and title blocks. This version keeps only clean product references and English product details. Final mechanical drawings should be issued as English-only engineering drawings before production approval.

8. User Operation Instructions - English Only

IR Programming

- Press and hold the Power button on this product for 3 seconds. When the indicator flashes slowly, the product enters IR programming mode.
- Place this product and the TV remote head-to-head on a flat surface about 1 inch apart. Press the Power button on the TV remote control.
- The indicator light will stop flashing when programming is complete.
- Power and Mute can be programmed by the same procedure.

Factory Reset

- Press and hold Power + Delete for 5 seconds. The indicator flashes to prompt restore to factory settings.

Cursor Speed

- Hold OK and press Volume-. The cursor speed becomes slower when the LED changes from blinking to solid.
- Hold OK and press Volume+. The cursor speed becomes faster when the LED changes from blinking to solid.

Backlight Function

- After the battery is installed, the backlight turns on automatically.
- Long press Menu to turn the backlight on or off.
- Backlight duration is 5 seconds. After 5 seconds, press any key to activate the backlight again.

Low Battery Reminder

- The red light flashes slowly when the AAA battery power is lower than 2 V. Replace the battery at this time.

9. Calibration, USB Receiver, and Bluetooth Pairing

Sensor Calibration

- Automatic calibration: when the device works, the cursor may drift due to voltage or temperature changes. Place the device flat on a desktop and keep it still. After the device enters quiescent mode, wake it again to complete calibration.
- Manual calibration: press and hold OK while installing the battery. After the red LED changes from flashing to normally on, place the product flat on a desktop and wait for the indicator to flash. Calibration is then complete.

USB Receiver Pairing

- For initial use, plug the USB receiver into the USB port of the device and wait 20 to 60 seconds for the USB receiver driver to install.
- Move the remote. If the cursor moves on the screen, pairing is successful.
- If pairing is unsuccessful, press and hold OK + Back until the LED starts flashing. Pairing is successful when the LED stops flashing.

Bluetooth Pairing and Deleting Pairing Information

- Before use, turn on the Bluetooth function of the main device.
- Press and hold OK + Return. When the LED flashes quickly, the product enters pairing mode.
- On the main device, add or search for a new device. Example device name in the source file: G20S PRO. Click the name to connect.
- After successful connection, the mouse pointer direction can be controlled by waving the remote in the air.
- If connection is unsuccessful, repeat the pairing steps.
- Typical reconnect time is 3 to 5 seconds after normal pairing with the main equipment.
- If the product connects to a new device, historical connection information must be deleted from the previous device before new search and pairing.

Operating Notes

- Some multimedia devices may not support the air mouse function. If OK does not work during device operation, close the air mouse pointer and try again.
- IR programming keys are available for many popular TV, voice device, and A/V receiver products, but they are not suitable for all brands or all items.
- Ensure the batteries have sufficient power. Low battery power can affect backlight brightness and air mouse cursor stability.
- The actual working space can be affected by electromagnetic fields. Ensure there is no strong electromagnetic interference in the working area.

10. Major Materials and Packaging Summary

Assembly	Major items retained in English-only summary
Remote control PCBA	FR4 2-layer PCB with OSP + carbon oil; HB9261 programmed IC; ME2135A33M6G DC-DC IC; ICM-42670-P sensor; 24 MHz crystal; capacitors, resistors, diodes, transistors, LEDs, inductor, IR transmitter
Mechanical parts	ABS top cover, laser-engraved bottom cover, ABS battery cover, transparent key bracket, light shield, silicone keypad, black-zinc screw, positive/negative/connecting springs
USB receiver	FR4 2-layer immersion-gold PCB; HB9266B programmed IC; 24 MHz crystal; USB metal assembly; ABS top and bottom shells; PE bag
Packaging	Outer label 100 mm x 75 mm; flat card 370 mm x 220 mm; carton 400 mm x 240 mm x 243 mm; export disinfected wood pallet 1000 mm x 1000 mm x 120 mm; protective corners
Pack quantity reference	Source packaging table indicates 88 pcs per carton and 3,520 pcs per pallet reference base

USB Dongle Visual Reference

Dongle：



Removed Content

Chinese-only cover sheets, Chinese annotations, duplicated bilingual instruction blocks, Chinese BOM tables, Chinese title blocks, and Chinese-labeled drawings were not carried over into this English-only specification. Any final production drawing set should be recreated with English labels before client approval.